Sprint Progress Report 6

Sprint Overview

Development of Website and Construction of Database.

In Sprint 6, we focused on closing the remaining MVP-level workflow gaps left after Sprint 5. The goal of this sprint was not to introduce new product areas, but to complete the existing user journey and administrative controls that were already implied by the requirements and use cases. Specifically, we concentrated on login-state closure, group join/leave behavior, settlement completion, reminders, and admin-facing account and audit capabilities.

This sprint aimed to convert the current application from a mostly connected expense-tracking flow into a more complete and reliable system. The login state is now handled more consistently across page navigation; group membership actions are exposed through a complete workflow; settlement records can be created and reflected in the ledger; reminders can be issued for outstanding balances; and administrators can review logs as well as disable or restore user accounts when needed.

Class Presentation

exhibition to peers.

The check-in process was maintained on a weekly basis to review the implementation status, validate the remaining requirements, and keep the final scope focused on the missing closing loops rather than introducing unrelated features.

Work Completed

Login Stage in Frontend and Backend

- The login flow now creates and preserves a consistent session-style user state after authentication.
- The frontend stores the authenticated user payload in `localStorage` and reuses it across pages that depend on the current user context.
- The logout action clears the stored session state and returns the application to an unauthenticated state.
- The login and register pages continue to link to the backend authentication endpoints, so the same user record is used throughout the application flow.

Group Join and Leave Workflow

- Group membership actions were completed so that a user can join a group through an invite-based or group-code-based workflow.
- Group owners can add and remove members through the backend workflow, and membership state is reflected consistently in the group data returned to the frontend.
- The leave-group behavior was aligned with the membership lifecycle so that users can exit a group when the workflow allows it.
- Group membership validation was kept strict so that membership actions do not silently create inconsistent records.

Settlement Closure

- Settlement records now support the complete payment-completion flow required by the sprint scope.

- Marking a payment as completed updates the settlement state and reflects the transaction in the group's ledger context.
- The backend settlement workflow now supports querying settlement records so that the current state of repayments can be reviewed.
- Balance updates remain consistent with the recorded settlement activity, which keeps the history and summary views aligned.

Reminder System

- The reminder flow was implemented to allow outstanding balances to be surfaced to the relevant participants.
- Reminder actions are associated with the correct group context so that notifications are not sent outside the intended membership scope.
- Reminder history can be retained for review, which helps confirm that the reminder workflow is operating as expected.

Admin View for Logs and Account Control

- Administrative visibility into audit and system logs was added so that key actions can be reviewed after they occur.
- User account control actions were completed so that an administrator can disable or restore user access when needed.
- These admin actions were written to preserve a clear record of the change, so the system can be audited later.
- The admin workflow remains intentionally limited to the required controls for this sprint and does not expand into unrelated platform management features.

Coverage of API and Validation

- The core API surface now includes the remaining endpoints needed for the sprint scope: login-state handling, group membership actions, settlement completion, reminder actions, and administrative controls.
- Validation was tightened around membership and settlement actions to reduce inconsistent writes.
- The automated test suite was expanded to cover the new flows and the main failure paths introduced by this sprint.
- Existing expense and history behavior continues to work alongside the new closing-loop features.

Work Division

For the sixth sprint, the team organized the remaining work into five focused implementation areas to keep the scope narrow and testable.

- Login-state closure, including logout and consistent authenticated user handling across pages.
- Group membership completion, including join, leave, add, and remove flows.
- Settlement completion, including payment completion and ledger consistency.
- Reminder handling for outstanding balances within the correct group scope.
- Administrative log review and account disable/restore behavior.

With this division, the sprint remained focused on the highest-value remaining MVP paths and avoided drifting into optional platform features.

Project Artifacts Updates

- The backend structure continues to use models, schemas, repositories, services, and routers, and Sprint 6 further completed the workflow coverage across those layers.
- The frontend pages for login, register, dashboard, history, and expense entry continue to work with the current backend behavior and authenticated user state.
- The automated test suite now covers the remaining workflow closure cases, including settlement, membership, and administrative actions.
- The database schema and ORM layer continue to represent the core MVP entities, including users, sessions, groups, group members, expenses, settlements, reminder history, and audit logs.

Weekly Meeting Summary

During the weekly meeting, we reviewed the remaining requirements and confirmed that the final sprint should focus on completion rather than expansion. The team agreed to prioritize the missing closure paths in the order that most directly affects the MVP user journey: login state, group membership lifecycle, settlement completion, reminder delivery, and administrative oversight.

We also discussed the trade-off between adding new functionality and finishing the existing one. The decision was to avoid scope creep and concentrate on the remaining use cases that were still incomplete, especially those involving repayment completion and admin review of system activity.

Challenges & Learnings

- One challenge was keeping the login state simple while still making it stable enough to persist across pages.
- Group membership actions required careful validation so that adding, removing, joining, and leaving a group would not create inconsistent records.
- Settlement completion needed to stay synchronized with the balance and history views, so the ledger could remain trustworthy.
- Admin visibility required a minimal but clear audit trail so that sensitive actions such as account disable or restore could be reviewed later.
- A key learning was that the remaining work was less about adding new screens and more about closing the workflows that were already implied by the existing requirements.

Next Steps

- Finalize any remaining edge cases in login-state handling.
- Recheck membership validation and ensure the join/leave workflow behaves consistently in all supported paths.
- Confirm that settlement completion updates the related balance and history data correctly.
- Review reminder behavior and ensure it stays constrained to the correct group context.
- Verify that admin log review and account control actions are recorded properly and remain auditable.
- Run the full automated test suite and perform a final browser smoke test on the main frontend pages.
- Prepare the project for final submission by checking that the remaining MVP requirements are covered and that no unrelated features were introduced.